

and timely. The integrity and accuracy of wholesale payment transactions depend on the use of proper control procedures throughout all phases of processing, including outsourced functions.

Regardless of whether the financial institution's control procedures are manual or automated, internal controls should address the areas of transaction initiation, data entry, computer processing, and distribution of output reports. Financial institutions should also maintain effective control over service provider access to customer and financial institution information consistent with GLBA 501(b). Contractual provisions should define the terms of acceptable access and potential liabilities in the event of fraud or processing errors. Refer to IT Handbook's Outsourcing Technology Services Booklet for more detail.

Endnotes

[1]	For further information on these topics, refer to the Committee on Payment and Settlement Systems (CPSS) "Recommendation for Central Counterparties," at www.bis.org .
[2]	FedLine® is a registered trademark of the Federal Reserve Banks.
[3]	In addition, FedwireFedwire® is a registered service mark of the Federal Reserve Banks. See http://www.frb services.org/ for further information on Fedwire Funds and Securities Service, and NSS.
[4]	CHIPS is a private multilateral settlement system operated by CHIP Co., a subsidiary of The Clearing House (formerly known as the New York Clearing House Association).
[5]	See Appendix D for a discussion of the general legal framework for interbank payment systems.
[6]	CHIP Co., using a formula based on the latest transaction history of each participant, establishes the amount of a participant's opening position requirement.
[7]	X.25 is the software that SWIFT provides to its users for access capability to SWIFT's core FIN application.
[8]	Government-sponsored enterprises (GSEs) are publicly-traded corporations created by Congress to address public policy concerns about the ability of members of certain groups to borrow sufficient funds at affordable rates. GSEs do not receive federal funds and rely primarily on debt financing for their day-to-day operations. GSE securities are not government securities; however, market participants rate and price GSE securities similar to U.S. government-issued securities.
[9]	Primary dealers are designated trading counterparties for the Federal Reserve Bank of New York in its execution of market operations to carry out US monetary policy. Currently there are 23 designated primary dealers.
[10]	Refer to http://www.frb services.org/Wholesale/ProtocolCertVendors.cfm for a list of protocol certified vendors.
[11]	See the IT Handbook's, "FedLine Booklet."
[12]	For a complete discussion of legal requirements, consult 31 CFR Part 500 et seq.
[13]	Chapter 53 of title 31, United States Code, the Bank Secrecy Act, and the implementing regulations promulgated by the Department of Treasury at 31 CFR part 103.
[14]	See Appendix F for a discussion of the "Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System."

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| [15] | See SR Letter 03-9, dated May 28, 2003, Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System, for further information. The agencies included the Board of Governors of the Federal Reserve System, the Office of the Comptroller of the Currency, and the Securities and Exchange Commission. |
| [16] | Federal Reserve Operating Circulars are available at http://www.frb.services.org/OperatingCirculars/index.html . |
| [17] | See http://www.federalreserve.gov/paymentsystems/psr/default.htm . |
| [18] | See IT Handbook "Retail Payment Systems Booklet" for additional information on NSS and PSR policy. |
| [19] | See SR Letter 03-9, dated May 28, 2003, Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System, for further information. The agencies included the Board of Governors of the Federal Reserve System, the Office of the Comptroller of the Currency, and the Securities and Exchange Commission. |

Appendix A: Examination Procedures

Tier I Examination Objectives and Procedures

EXAMINATION OBJECTIVE: Examiners should use the Wholesale Payment Systems Examination Procedures to determine the adequacy of the financial institution's payment system risk policies and wholesale payment business processes, including personnel and internal control systems used to mitigate the risks associated with wholesale payment systems. Wholesale payment system services include Fedwire Funds Service funds transfer and book-entry securities; CHIPS; SWIFT; payment messaging systems; net settlement, clearing and settlement systems; internally developed and off-the-shelf funds transfer systems; and web-based payment systems. The examiner's assessment of risk and risk management practices relating to a financial institution's wholesale payment system service should help determine the extent of testing and which procedures to perform. The assessment should consider the effectiveness of formal policies and procedures as well as the financial institution's underlying internal control environment including information security, business continuity and disaster recovery, and management of wholesale payment services outsourced to third parties.

Financial institutions are exposed to numerous credit, liquidity, reputation, legal, and operational risks in provisioning wholesale payment system services to counter parties and performing related processing, clearance, and settlement functions in-house and with third parties. Depending on the financial risks, IT related operational (transactional) risks, compliance risks, and complexity of wholesale payment system activity, the examination may require an integrated team approach that includes the knowledge and skills of safety and soundness examiners and IT examiners.

Examiners may incorporate the Examination Procedures as part of either an IT or safety and soundness examination. The Examination Procedures can also be used in its entirety, or can be used in modular fashion, focusing on particular wholesale payment system products or business lines. Depending on the size and complexity of the financial institution or service provider, examiners may tailor the use of the examination procedures. In many cases, they can eliminate certain procedures and still arrive at a conclusion regarding the quality of risk management practices and performance. The examination procedures are structured as follows:

- Tier I objectives and procedures, which evaluate the effectiveness of the financial institution and service provider's wholesale payment systems, internal controls, and risk management processes that may be relied on for the purpose of identifying and managing risks.
- Tier II objectives and procedures, which provide additional validation as warranted by the risks to verify the effectiveness of the financial institution and service provider's wholesale payment systems function.

Objective 1: Determine the scope and objectives of the examination of the wholesale

payment systems function.

1. Review past reports for comments relating to wholesale payment systems. Consider:

- Regulatory reports of examination.
- Internal and external audit reports.
- Regulatory reports on and, audit, and information security reports from/on service providers.
- Trade group, card association, interchange, and clearing house documentation relating to services provided by the financial institution.
- Supervisory strategy documents, including risk assessments.
- Examination work papers.

2. Review past reports for comments relating to the institution's internal control environment and technical infrastructure. Consider:

- Internal controls including logical access controls, data center operations, and physical security controls.
- Wholesale EFT network controls.
- Inventory of computer hardware, software, and telecommunications protocols used to support wholesale EFT transaction processing.

3. During discussions with financial institution and service provider management:

- Obtain a thorough description of the wholesale payment system activities performed, including transaction volumes, transaction dollar amounts, and scope of operations, including Fedwire Funds Service, CHIPS, SWIFT, and all wholesale payment messaging systems in use.
- Review the financial institution's payment system risk policy and evaluate its compliance with net debit caps and other internally generated self-assessment factors.
- Identify any wholesale payment system functions performed via outsourcing relationships and determine the financial institution's level of reliance on those services.
- Identify any significant changes in wholesale payment system policies, personnel, products, and services since the last examination.

4. Review the financial institution's response to any wholesale payment systems issues raised at the last examination. Consider:

- Adequacy and timing of corrective action.
- Resolution of root causes rather than specific issues.
- Existence of outstanding issues.

Objective 2: Determine the quality of oversight and support provided by the board of directors and management.

1. Determine the quality and effectiveness of the financial institution's wholesale payment systems management function. Consider:

- Data center and network controls over backbone networks and connectivity to counter parties.
- Departmental controls, including separation of duties and dual control procedures, for funds transfer, clearance, and settlement activities.
- Compliance with the Federal Reserve's Payment System Risk policies and procedures.
- Physical and logical security controls designed to ensure the authenticity, integrity, and confidentiality of wholesale payments transactions.

2. Assess management's ability to manage outsourcing relationships with service providers and software vendors contracted to provide wholesale payment system services. Evaluate the adequacy of terms and conditions, and whether they ensure each party's liabilities and responsibilities are clearly defined. Consider:

- Adequacy of contract provisions including service level and performance agreements.
- Compliance with applicable financial institution and third party (e.g. Federal Reserve, CHIPS, SWIFT) requirements.
- Adequacy of contract provisions for personnel, equipment, and related services.

3. Evaluate the adequacy and effectiveness of financial institution and service provider contingency and business recovery plans. Consider:

- Ability to recover transaction data and supporting books and records based on wholesale payment system business line requirements.
- Ability to return to normal operations once the contingency condition is over.
- Confidentiality and integrity of interbank and counter party data in transit and storage.

4. Evaluate wholesale payment system business line staff. Consider:

- Adequacy of staff resources.
- Hiring practices.
- Effective policies and procedures outlining department duties.
- Adequacy of accounting and financial controls over wholesale payment processing, clearance, and settlement activity.

5. Review the disaster recovery plan for the funds transfer system (FTS) to ensure it is reasonable in relation to the volume of activity, all units of the FTS are provided for in the plan, and the plan is regularly tested.

Objective 3: Determine the quality of risk management and support for Payment System Risk policy compliance.

1. Review policies and procedures in place to monitor customer balances for outgoing payments to ensure payments are made against collected funds or established intraday or overnight overdraft limits and payments resulting in excesses of established uncollected or overdraft limits are properly authorized.

2. Review a sample of contracts authorizing the institution to make payments from customers' accounts to ensure they adequately set forth responsibilities of the institution and the customer, primarily regarding provisions of the Uniform Commercial Code Article 4A (UCC4A) related to authenticity and timing of transfer requests.

Objective 4: Determine the quality of risk management and support for internal audit and the effectiveness of the internal audit program for wholesale payment systems.

1. Review the audit program to ensure all functions of the FTS are covered. Consider:

- Payment order origination (funds transfer requests).
- Message testing.

- Customer agreements.
- Payment processing and accounting.
- Personnel policies.
- Physical and data security.
- Contingency plans.
- Credit evaluation and approval.
- Incoming funds transfers.
- Federal Reserve's Payment Systems Risk Policy.

2. Review a sufficient sample of supporting audit work papers necessary to confirm that they support the execution of procedures established in step 1 above.

3. Review all audit reports related to the FTS and determine the current status of any exceptions noted in the audit report.

CONCLUSIONS

1. Determine the need to proceed to Tier II procedures for additional validation to support conclusions related to any of the Tier I objectives.

2. From the procedures performed, including any Tier II procedures performed:

- Document conclusions related to the quality and effectiveness of the retail payment systems function.
- Determine and document to what extent, if any, the examiner may rely upon wholesale payment systems procedures performed by internal or external audit.

3. Review your preliminary conclusions with the EIC regarding:

- Violations of law, rulings, regulations, and third party agreements.
- Significant issues warranting inclusion as matters requiring board attention or recommendations in the report of examination.

- Potential impact of your conclusions on URSIT composite and component ratings.

4. Document your conclusions in a memo to the EIC that provides report ready comments for all relevant sections of the FFIEC Report of Examination and guidance to future examiners.

5. Organize work papers to ensure clear support for significant findings and conclusions.

Tier II Examination Objectives and Procedures

Overall Objective: The Tier II examination procedures for Wholesale Payment Systems provide for additional verification procedures to evaluate the effectiveness of the financial institution's internal control processes over its wholesale payment systems, including Fedwire Funds Service funds transfer and book entry securities, CHIPS, SWIFT, payment messaging systems, net settlement, clearing and settlement systems, internally developed and off-the-shelf funds transfer systems, and web-based payment systems. These procedures are designed to assist in achieving examination objectives, and may be used in their entirety or selectively. Examiners should coordinate this coverage with other examiners involved in assessing the institution's information systems, operations, and information security effectiveness to ensure there is an adequate understanding of the control environment as it pertains to the bank's wholesale payment systems.

Objective 1: Determine if management and the board have enacted sufficient controls over funds transfer activity.

1. Determine if management and the board provide administrative direction for the funds transfer function. Ascertain whether:

- The directors and senior management are informed regarding the nature and magnitude of risks with the institution's funds transfer activities.
- Management is informed of new systems designs and available hardware for the wire transfer system.
- The board of directors and/or senior management regularly review and approve any funds transfer limits, and if so, when the limits were last reviewed.
- Senior management and the board monitor customers with large intraday or overnight overdrafts and analyze the overdrafts along with all other credit exposure to the customer.

2. Determine if the board and management have developed sufficient policies and procedures to ensure that the following are reviewed:

- Transaction volumes.
- Adequacy of personnel and equipment.
- Customer creditworthiness.
- Funds transfer risk.

3. Determine if the board and senior management develop and support adequate user access procedures and controls for funds transfer requests. Assess whether the institution:

- Maintains a current list of employees approved to initiate funds transfer requests.
- Has developed and approved an organization plan that shows the structure of the funds management department and limits the number of employees who can initiate or authorize transfer requests.
- Has a list of authorized employee signatures maintained in a secure environment.
- Regularly reviews staff compliance with credit and personnel procedures, operating instructions, and internal controls.
- Requires its senior management receive and review activity and quality control reports which disclose unusual or unauthorized activities and access attempts

4. Determine if management maintains authorization lists from its customers that use the funds transfer system. Verify:

- Management advises customers to limit the number of authorized signers.
- There are dual controls or other protections over customer signature records.
- The authorization list also identifies authorized sources of requests (e.g., telephone, fax, memo, etc.).
- The customer authorization establishes limits over the amount each signer is authorized to transfer.

5. Determine if the institution has dual control procedures that prohibit persons who receive transfer requests from transmitting or accounting for those requests.